

The quadratic formula

WORKSHEET



Interactive version

Name

Class

Date

REMEMBER

Coefficients and square roots

Level 1 • threshold

1.

Write $2x^2 = 7x + 4$ in the form $ax^2 + bx + c = 0$ with $a > 0$. Enter a , b and c , in this order, separated by semicolons.

Answer:

Level 1 • threshold

2.

Work out $\sqrt{9 + 16}$.

Answer:

WORKED EXAMPLES

Worked examples

YOUR TURN

Level 1 • threshold

Solve $x^2 + 3x - 4 = 0$ with the formula.

Answer:

Example 2 • When $a \neq 1$: $2x^2 - 3x - 2 = 0$

COEFFICIENTS

$a = 2$, $b = -3$, $c = -2$. So $-b = 3$.

DISCRIMINANT

$D = (-3)^2 - 4 \cdot 2 \cdot (-2) = 9 + 16 = 25$, and $\sqrt{25} = 5$.

FORMULA • YOUR TURN

CHECK • YOUR TURN

YOUR TURN

Level 2 · satisfactory

Solve $3x^2 - 5x - 2 = 0$ with the formula.

Answer:

CHECK

Check yourself

Level 1 · threshold

1.

What is the discriminant of $3x^2 - 7x + 2 = 0$?

☐ A -73

☐ B 43

☐ C 25

☐ D 73

Level 1 · threshold

2.

Solve $2x^2 + x - 6 = 0$.

Answer:

Level 2 · satisfactory

3.

Tomas solved the same equation, $2x^2 + x - 6 = 0$, and got $x = 2$ and $x = -\frac{3}{2}$. Check his first root: substitute $x = 2$ into the left side $2x^2 + x - 6$. What number do you get?

Answer:

4.

Here are the lines of the proof, without the right-hand column. For each step (1)–(5), write what was done and why it is allowed.

$$ax^2 + bx + c = 0 \quad | \quad (1)$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad | \quad (2)$$

$$x^2 + \frac{b}{a}x = -\frac{c}{a} \quad | \quad (3)$$

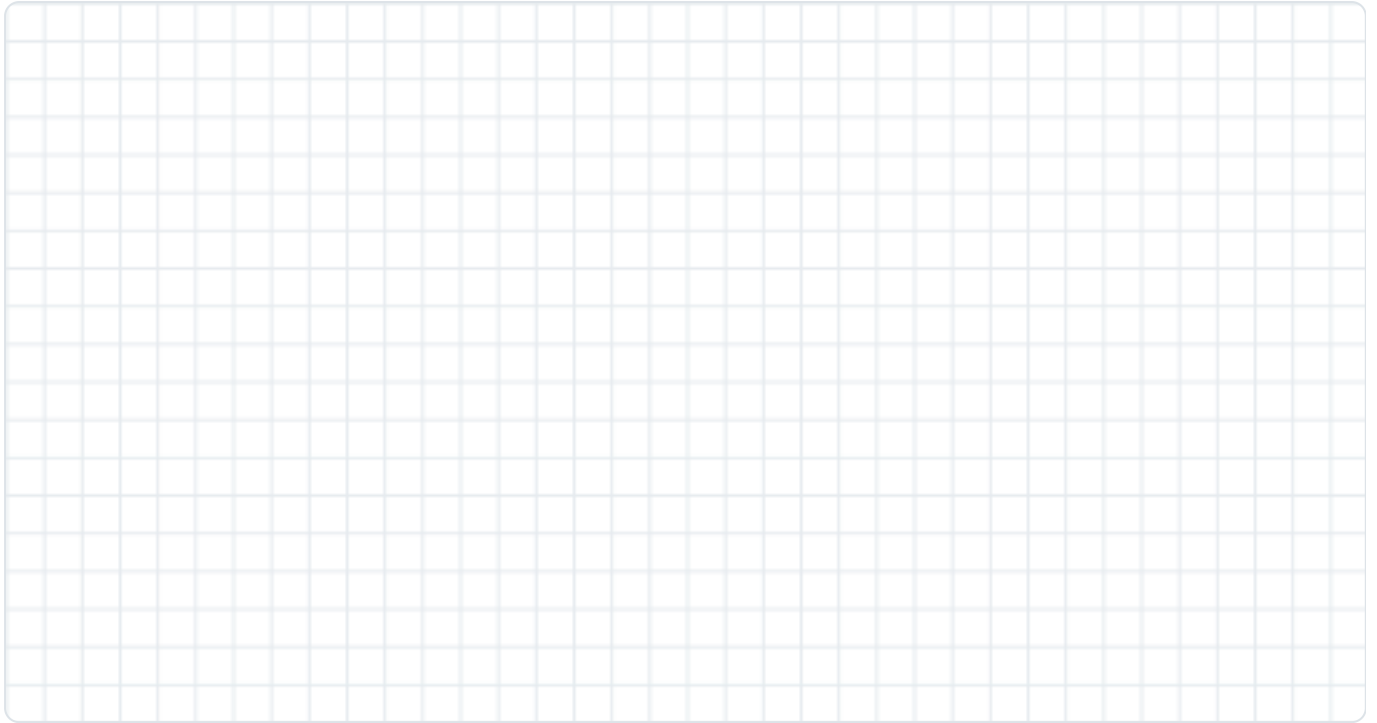
$$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a} \quad | \quad (4)$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \quad | \quad (5)$$

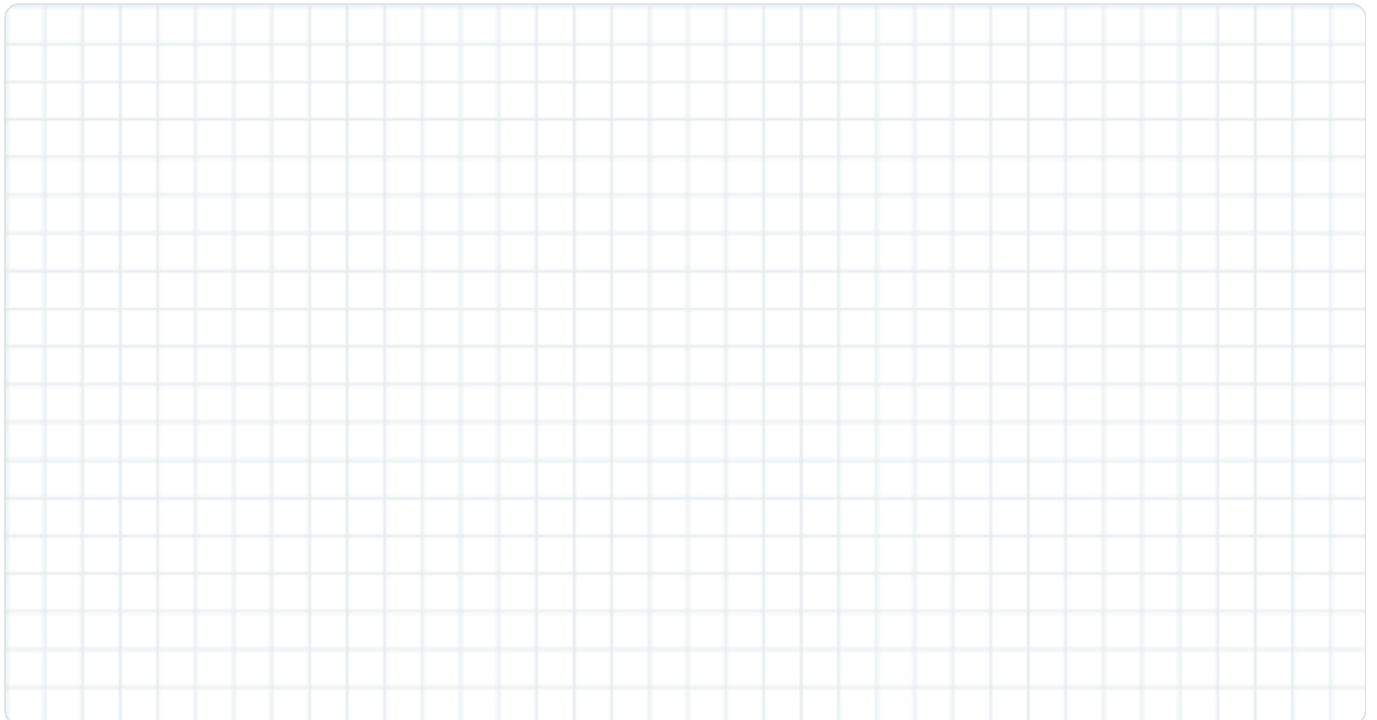
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

5.

Solve the equation $x^2 + 2x - 15 = 0$. Show how you find the discriminant, then find both roots.

**6.**

Prove the quadratic formula yourself. Start from $ax^2 + bx + c = 0$ with $a \neq 0$, make the four moves, and give a reason for every step.



EXAM TASK

Two consecutive integers

Level 3 · basic

PUPP style · 4 points

The sum of the squares of two consecutive positive integers is 181.

- a) Call the smaller integer n . Write an equation for n .
- b) Show that the equation can be written as $n^2 + n - 90 = 0$.
- c) Find the two integers.

